

EduQoo — Architecture

Technical reference for the Android application

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Technical reference for the Android application. This document describes structure and behaviour only. Product rationale is out of scope.

- **Target:** Android 13+, single flagship device (iQOO loaner).
- **Language:** Kotlin. UI in Jetpack Compose.
- **Network dependency:** none required at runtime.

1. System overview

The application is a single Android process. It contains five layers.

+-----+		
1. UI	Jetpack Compose	
	TutorScreen · TeacherScreen · Trace	
+-----+		
2. Orchestration	EduQooCore	
	router · budgeter · planner · verifier	
+-----+		
3. Tools	7 registered functions, typed	
	ink · solve · speak · retrieve · class	
+-----+		
4. Engines	MediaPipe LLM · ML Kit · Symbolic ·	
	Android TTS · CameraX · Ktor	
+-----+		
5. Storage	SQLite (Room) · app-private files	
+-----+		

No layer calls upward. Layer 2 is the only layer permitted to invoke layer 3, and layer 3 is the only layer permitted to invoke layer 4.

2. Modules

MODULE	TYPE	RESPONSIBILITY
:app	Android application	Compose UI, navigation, permissions, lifecycle
:core	Kotlin library	EduQooCore — routing, context assembly, planner loop, verifier
:tools	Kotlin library	Tool definitions and their JSON Schema projections
:inference	Kotlin library	MediaPipe LLM Inference wrapper; session lifecycle
:ink	Kotlin library	CameraX capture, ML Kit recognition, line segmentation
:solver	Kotlin library	Expression parser and equivalence checker
:speech	Kotlin library	TextToSpeech wrapper, language + voice-pack management
:server	Kotlin library	Ktor CIO server, hotspot, mDNS, SSE broadcast
:data	Kotlin library	Room entities, DAOs, vector store, CSV export

:core depends on :tools only. :tools depends on the engine modules. The engine modules depend on nothing internal. This prevents circular dependencies and allows each engine to be unit-tested without the runtime.

3. The tutor pipeline

The primary data flow. All stages execute on-device.



`InkSource` is either `Photo(Bitmap)` → ML Kit Text Recognition v2, or `Strokes(Ink)` → ML Kit Digital Ink Recognition. Lines are ordered top to bottom by `bounds.top`.

Confirm — `:app` If any `RawLine.confidence < CONFIRM_THRESHOLD` (0.90), the pipeline suspends and the confirm sheet is shown. The pipeline resumes only with a user-acknowledged `List<String>`. This gate is unconditional: it cannot be skipped by configuration.

Parse — `:solver`

```
sealed interface ParseResult { data class Ok(val expr: Expr) ; data class Failed(val at: Int) }
fun parse(line: String): ParseResult
```

A parse failure aborts the pipeline with a user-facing message. It is never passed to the model to "interpret".

Equivalence check — `:solver`

```
data class StepResult(val index: Int, val equivalent: Boolean, val residual: String)
fun check(steps: List<Expr>): List<StepResult>
```

For each `i > 0`, the checker evaluates whether `steps[i]` and `steps[i-1]` are equal by normalising `steps[i] - steps[i-1]` (equations are first rewritten to `lhs - rhs`) and testing the result against zero. The first index with `equivalent == false` is the reported error. Later steps are still checked and reported, but not surfaced as the primary error.

The output is deterministic and does not involve the language model.

Narrate — `:core` → `:inference`

```
suspend fun narrate(trace: List<SolveStep>, lang: Lang): List<String>
```

The model receives the solver's own trace and produces at most one sentence per step, bounded at 25 words. It is not given the student's incorrect working as authoritative input, and it is not asked to determine correctness.

Speak — `:speech`

```
suspend fun speak(sentences: List<String>, lang: Lang)
```

`TextToSpeech.synthesizeToFile` per sentence, then sequential playback so the audio can be paused and stepped alongside the animation.

4. EduQooCore

The orchestration layer in `:core`.

4.1 Execution



Passes 1 and 2 are separate `LlmInference` calls with different configurations. They are not merged: enabling schema constraints and tool calling in a single decode makes tool-call tokens unreachable under the grammar mask, and the model silently stops emitting them.

4.2 Context budget

SEGMENT	BUDGET	POLICY
System prompt	180	Static
Core memory	120	Pinned: grade, language, three weakest concepts
Retrieved	640	Top-k chunks, k = 4, current source only
Working	900	Conversation; compacted to a summary when 80% full
Scratchpad	0	Written to a file, re-read on demand
Total	2048	Hard ceiling

Raw tool output is dropped from working memory once consumed. Remaining capacity is passed back to the planner after each tool call so it can stop early rather than overflow.

4.3 Verifier

EduQoo — Architecturegithub.com/Harsh4-Dev/iqoo

CLAIM TYPE	CHECK	ON FAILURE
Mathematical	:solver re-evaluation	Discard and regenerate from the trace
Factual	Substring/embedding match against retrieved chunk	Replace with a refusal
Structural	JSON Schema validation	Retry pass 2, max 2 attempts

Nothing is rendered to a student before the verifier returns.

5. Tool registry

Seven tools. Each is a Kotlin function with a typed signature; the schema projection is generated at build time by a KSP processor, so the declaration and the schema cannot diverge.

TOOL	SIGNATURE	ENGINE
ink.read	(InkSource) -> List<RawLine>	ML Kit
solve.check	(List<Expr>) -> List<StepResult>	:solver
solve.steps	(Problem) -> List<SolveStep>	:solver
speak	(List<String>, Lang) -> Unit	Android TTS
retrieve	(String, Int) -> List<Chunk>	SQLite
class.broadcast	(Artifact, Audience) -> Receipts	Ktor SSE
progress.write	(StudentId, Concept, Outcome) -> Unit	Room

Tools are deliberately non-overlapping. Two tools with similar descriptions cause a small model to spend turns choosing between them.

6. Inference

PROPERTY	VALUE
Model	Gemma-2-2B, instruction-tuned
Quantisation	4-bit
Runtime	MediaPipe LLM Inference

PROPERTY	VALUE
EduQoo — Architecture github.com/Harsh4-Dev/iqoo	
Accelerator	Snapdragon NPU where the delegate is available, GPU otherwise
Distribution	Weights bundled in the APK / installed on first run to app-private storage
Session	One <code>LlmInference</code> instance, created in <code>onCreate</code> of the foreground service, kept warm
Concurrency	One inference at a time, enforced by a <code>Mutex</code> in <code>:inference</code>

Thermal policy. Only one generation may be in flight. There is no live camera analysis loop and no speculative pre-generation. A `PowerManager.OnThermalStatusChangeListener` raises the tier at `THERMAL_STATUS_MODERATE` : narration shortens to one sentence total, and at `THERMAL_STATUS_SEVERE` narration is skipped and only the solver output is shown.

7. Storage

Room, one database, app-private. No content provider is exported.

```

student(id, name, grade, lang)
attempt(id, student_id, captured_at, image_path, tier)
step(id, attempt_id, idx, raw_text, confidence, parsed, equivalent)
concept(id, slug, label)
observation(id, student_id, concept_id, outcome, observed_at)
source(id, title, page_count, indexed_at)
chunk(id, source_id, ord, text, page)
chunk_vec(chunk_id, embedding BLOB)          -- 768-dim, float32
chunk_fts(chunk_id, text)                     -- FTS5

```

Retrieval merges a vector scan over `chunk_vec` with an FTS5 match over `chunk_fts` . Vector search alone fails on exact terms — a formula name, a proper noun — which is why both exist.

Captured images are written to app-private storage and are never uploaded. They are deleted when their `attempt` row is deleted.

8. Classroom server

Runs only while a class is active.

CONCERN	IMPLEMENTATION
Server	Ktor CIO, bound to <code>0.0.0.0:8080</code>

CONCERN	IMPLEMENTATION
EduQoo — Architecture	github.com/Harsh4-Dev/iqoo
Lifecycle	Foreground service with a persistent notification
Access point	<code>WifiManager.startLocalOnlyHotspot()</code> ; SSID and passphrase encoded into a QR
Discovery	<code>NsdManager</code> advertising <code>_eduqoo._tcp</code> , resolving <code>class.local</code>
Client	Static HTML/JS served from <code>assets/</code>
Push	Server-Sent Events to all connected clients
Receipts	Client ACKs on render; the teacher UI shows delivered / pending
Integrity	SHA-256 per artifact; re-push dedupes on hash

`startLocalOnlyHotspot()` behaviour is OEM-dependent. If it fails, the server falls back to the device's current Wi-Fi network and relies on mDNS alone. Both paths are exercised in tests.

8.1 Routes

METHOD	PATH	PURPOSE
GET	<code>/</code>	Student client
GET	<code>/events</code>	SSE stream
POST	<code>/answer</code>	Quiz submission
GET	<code>/teacher/report.csv</code>	Class diagnostic export
POST	<code>/v1/chat/completions</code>	OpenAI-compatible endpoint over the LAN

9. Threading

WORK	DISPATCHER
Compose UI	Main
Camera capture	CameraX executor
ML Kit recognition	<code>Dispatchers.Default</code>
Parsing and equivalence	<code>Dispatchers.Default</code>
Inference	Dedicated single-thread dispatcher, guarded by <code>Mutex</code>
Room and file IO	<code>Dispatchers.IO</code>

WORK	DISPATCHER
EduQoo — Architecture Ktor	Its own CIO engine dispatcher

github.com/Harsh4-Dev/iqoo

Every long-running operation is a cancellable `suspend` function scoped to the screen's `viewModelScope`. Navigating away cancels in-flight inference.

10. Error handling

FAILURE	BEHAVIOUR
Camera unavailable	Fall back to the math keyboard input path
Recognition returns nothing	Show "we couldn't read the page"; offer keyboard
Any line below confidence threshold	Confirm sheet, pipeline suspended
Parse failure	Abort with the failing line highlighted; no model fallback
Solver timeout (> 3s)	Report unverified; do not display a correction
Inference timeout (> 20s)	Show solver output without narration
TTS voice pack missing	Show text explanation, surface a one-time download prompt
Hotspot creation fails	Fall back to the current network + mDNS

The general rule: when a deterministic stage fails, the pipeline stops. The language model is never used as a recovery path for a failed solver or parser.

11. Connectivity tiers

`TierMonitor` derives a tier from `ConnectivityManager` capabilities and observed throughput.

TIER	CONDITION	BEHAVIOUR
T0	No transport	All features complete. No queue drain.
T1	Transport, bandwidth < 200 kbps or high loss	Deltas queue; media deferred; UI never blocks
T2	Transport, bandwidth ≥ 200 kbps	Burst sync; cloud escalation permitted if confidence < 0.70

Tier affects synchronisation only. No tutor pipeline stage has a network dependency, so tier never changes tutor behaviour or output.

12. Dependencies

DEPENDENCY	PURPOSE	LICENCE
MediaPipe Tasks GenAI	LLM inference	Apache 2.0
ML Kit Text Recognition v2	Printed / photographed text	Apache 2.0 (bundled model)
ML Kit Digital Ink Recognition	Stroke input	Apache 2.0
CameraX	Capture	Apache 2.0
Ktor CIO	Embedded server	Apache 2.0
Room	Persistence	Apache 2.0
Symbolic solver	Equivalence checking	Verify before use
Jetpack Compose	UI	Apache 2.0

The solver licence must be confirmed against the exact release being linked. A restrictive licence discovered late is unrecoverable, because the solver is not replaceable by the model.

13. Build order

Each row depends on the rows above it.

#	COMPONENT	BLOCKING REASON
1	<code>:inference</code> wrapper	Everything downstream is a call into it
2	<code>:tools</code> registry + schema generation	The planner cannot run without typed tools
3	<code>:core</code> two-pass executor	Nothing is reliable until tool calls round-trip
4	<code>:solver</code> parse + equivalence	The product's only source of truth
5	<code>:ink</code> capture + recognition + confirm	The input path
6	Tutor UI + red-boundary overlay	The demo
7	<code>:speech</code> narration	Adds the language layer
8	<code>:data</code> observations + CSV export	Teacher analytics
9	<code>:server</code> hotspot + client	Multi-device

Steps 1–6 constitute the MVP. Steps 7–9 are additive and each is independently droppable.

14. Known constraints

1. **ML Kit has no Tamil script recognition model.** Tamil is an output language only. The recognised content is digits, Latin letters and mathematical symbols.
2. **Handwritten fraction and superscript recognition is unreliable.** The confirm gate exists because of this, and the math keyboard remains a first-class input path rather than a fallback of last resort.
3. `startLocalOnlyHotspot()` **is OEM-sensitive** and unverified on OriginOS. The mDNS fallback path must be tested on the target device before it is relied on.
4. **Sustained inference raises device temperature.** The thermal policy in §6 is required, not optional.
5. **Model download size affects first-run time.** If weights are not bundled, the first launch requires a network and the offline claim does not hold until it completes.

15. Prototype site

This repository also contains a browser prototype of the above, used for design and rehearsal. It is not the Android application and shares no code with it.

PATH	CONTENTS
<code>index.html</code>	Single-page prototype and design rationale
<code>assets/css/</code>	<code>tokens</code> · <code>site</code> · <code>phone</code> · <code>doodle</code>
<code>assets/js/data.js</code>	Fixture corpus: problems, scan output, chapter chunks, roster
<code>assets/js/app.js</code>	Prototype state machine, screen renderers, router, trace inspector
<code>assets/js/doodles.js</code>	Generated SVG scenes, one per screen
<code>assets/js/icons.js</code>	Icon set
<code>assets/js/qr.js</code>	Pre-generated QR matrix
<code>deck/</code>	Presentation deck, 1920×1080, for screen recording

The prototype is driven by scripted state and fixed timings. It performs no inference. Every figure it displays comes from `data.js`.